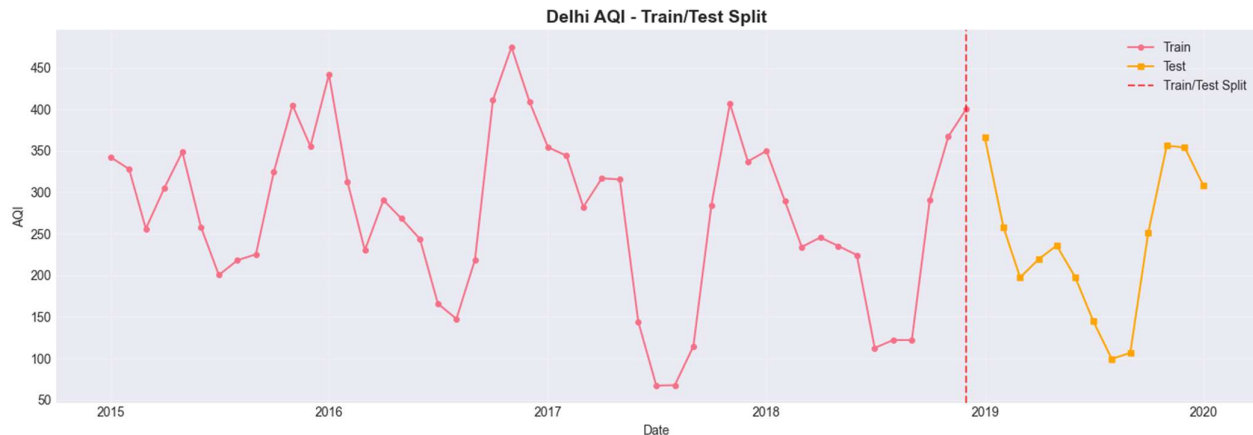


Test Report : Air Quality Analysis and Forecasting



SARIMA Model testing

SARIMA Model Summary:

SARIMAX Results

```
=====
Dep. Variable:          AQI      No. Observations:      48
Model:                 SARIMAX(0, 1, 3)x(0, 1, [1], 12)  Log Likelihood      -185.444
Date:                  Mon, 08 Dec 2025                AIC              380.888
Time:                  21:02:12                        BIC              388.665
Sample:                01-01-2015                      HQIC             383.573
                   - 12-01-2018
```

Covariance Type: opg

```
=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
ma.L1         -0.5591      8.143     -0.069      0.945     -16.519      15.401
ma.L2         -0.0814      3.616     -0.022      0.982      -7.169       7.006
ma.L3         -0.3558      3.058     -0.116      0.907      -6.349       5.638
ma.S.L12      -0.9978     121.732     -0.008      0.993     -239.588     237.593
sigma2        1325.9019    1.61e+05     0.008      0.993     -3.13e+05     3.16e+05
=====
```

```
=====
Ljung-Box (L1) (Q):      0.63  Jarque-Bera (JB):      0.24
Prob(Q):                 0.43  Prob(JB):          0.89
Heteroskedasticity (H):  0.54  Skew:             0.20
Prob(H) (two-sided):     0.29  Kurtosis:         3.01
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

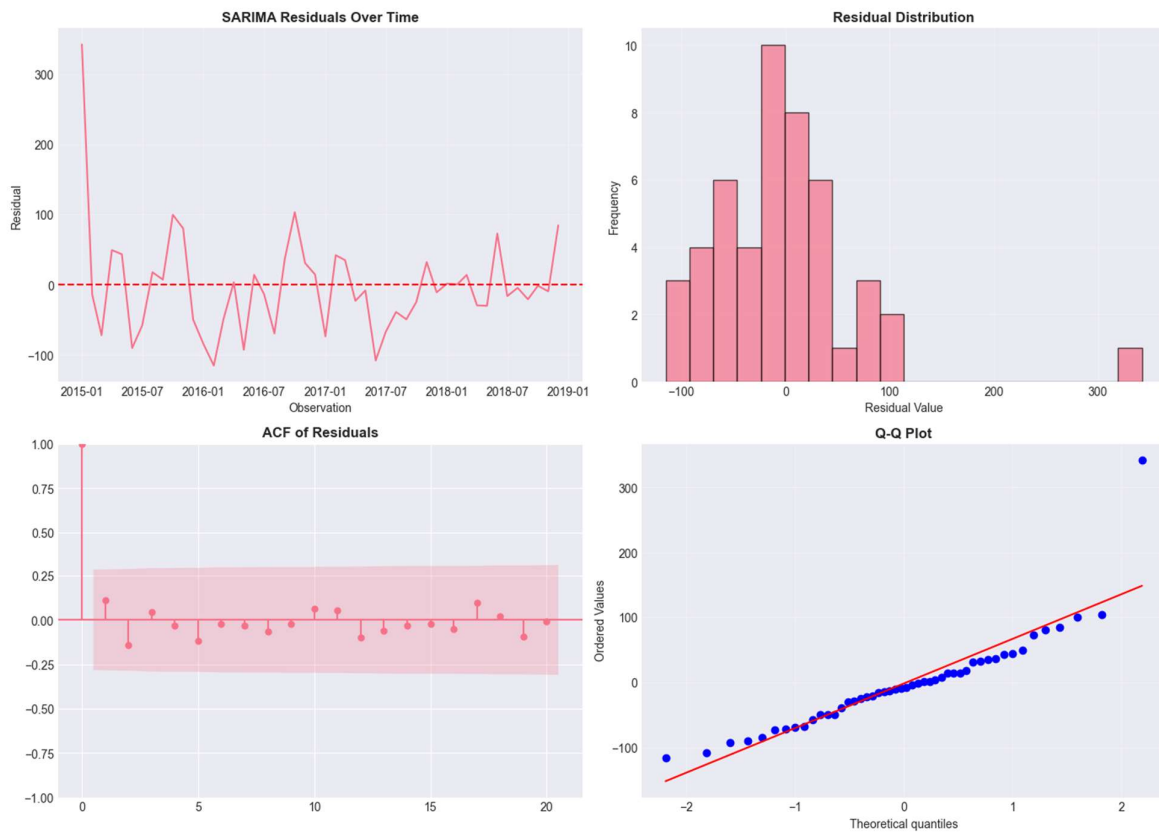
=== SARIMA Performance ===

Train RMSE: 66.39

Train MAE: 47.57

Test RMSE: 26.88 | Test MAE: 22.77 | Test MAPE: 11.14%

Residual diagnostics

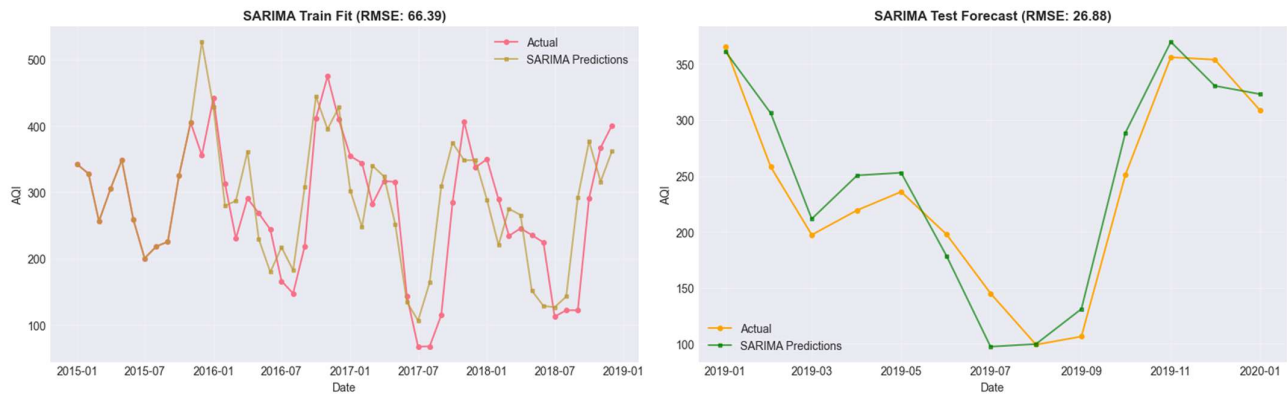


Ljung-Box Test (H0: residuals are independent):-

	lb_stat	lb_pvalue
10	3.275861	0.974167
20	6.139254	0.998696

Interpretation: p-value > 0.05 suggests residuals are independent (good)

=== SARIMA forecast visualization ===



=== LSTM Model Testing ===

Sequence length: 12 months

Number of training sequences: 36

Model: "LSTM-AQI-Forecaster"

```
# Build LSTM model
```

```
lstm_model = Sequential([  
    LSTM(32, activation='relu', return_sequences=True,  
input_shape=(n_input, n_features), name='LSTM_1'),  
    LSTM(32, activation='relu', name='LSTM_2'),  
    Dense(1, name='Output')  
], name='LSTM-AQI-Forecaster')
```

```
lstm_model.compile(optimizer='adam', loss='mse')
```

```
lstm_model.summary()
```

=== LSTM Training Loss Curve ===



Final training loss: 0.021013

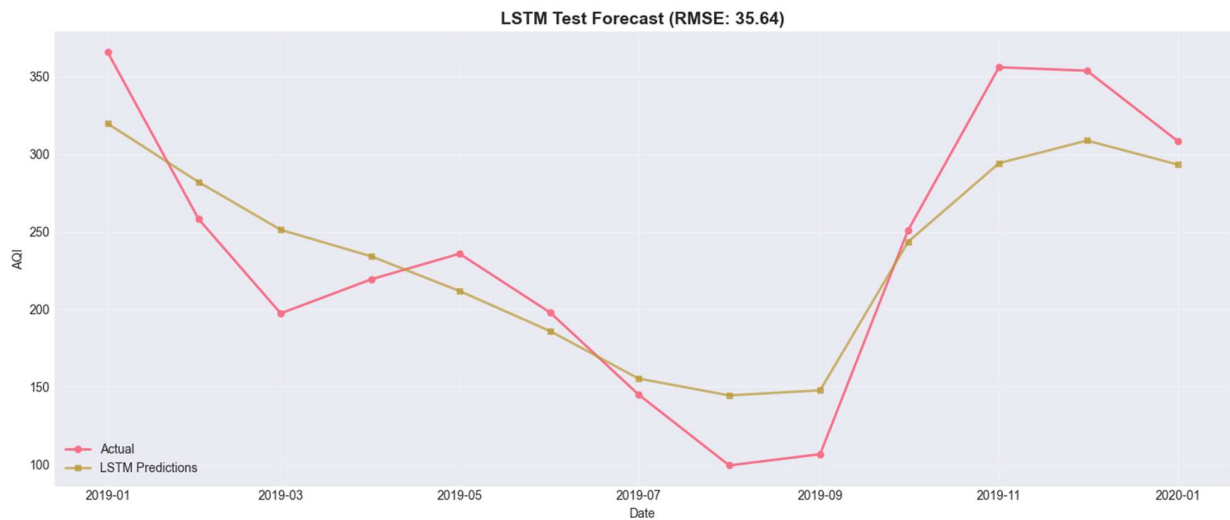
=== LSTM Performance ===

Test RMSE: 35.64

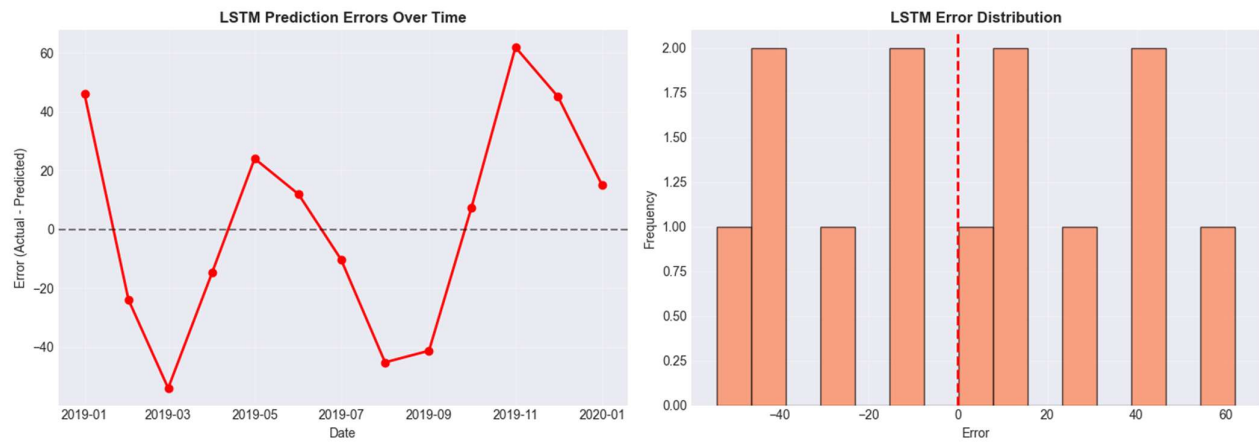
Test MAE: 30.82

Test MAPE: 15.48%

=== LSTM forecast visualization ===



=== LSTM prediction errors ===



Mean error: 1.73

Std dev of errors: 35.60

=== Prophet model testing ===

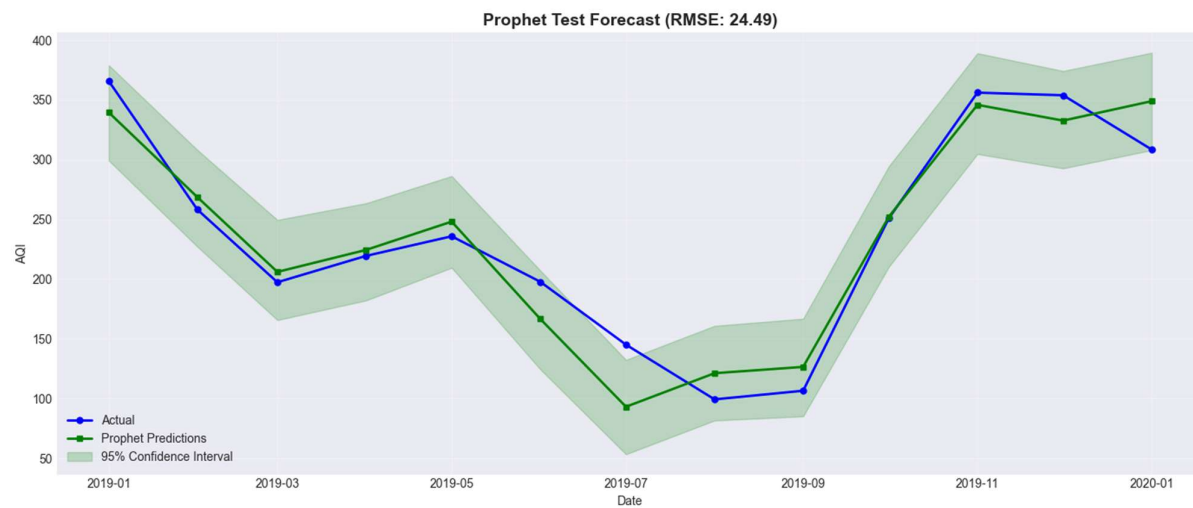
=== Prophet Performance ===

Test RMSE: 24.49

Test MAE: 20.05

Test MAPE: 10.61%

=== Prophet forecast visualization ===



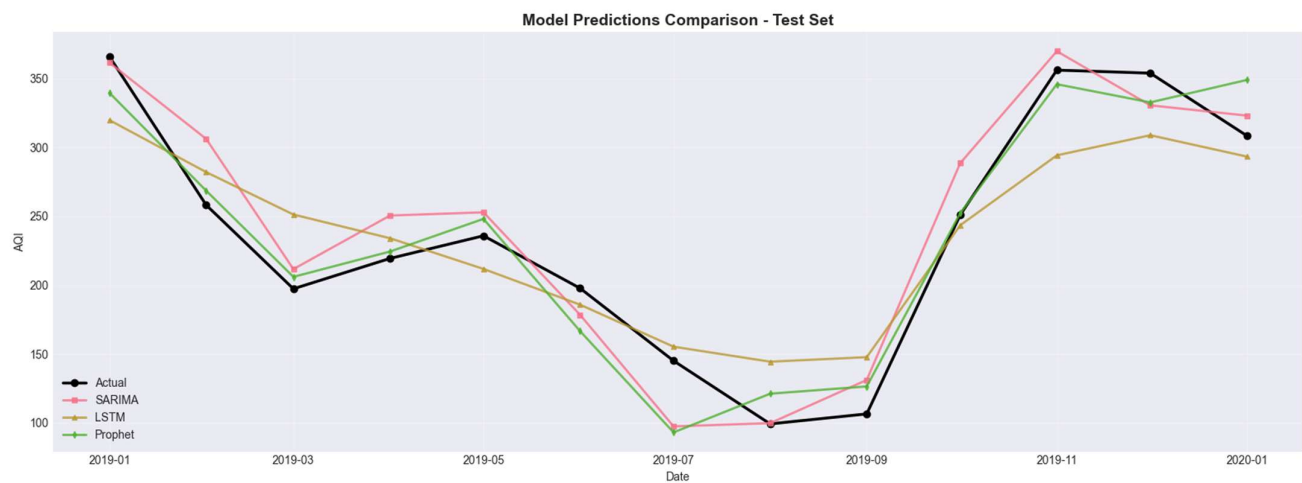
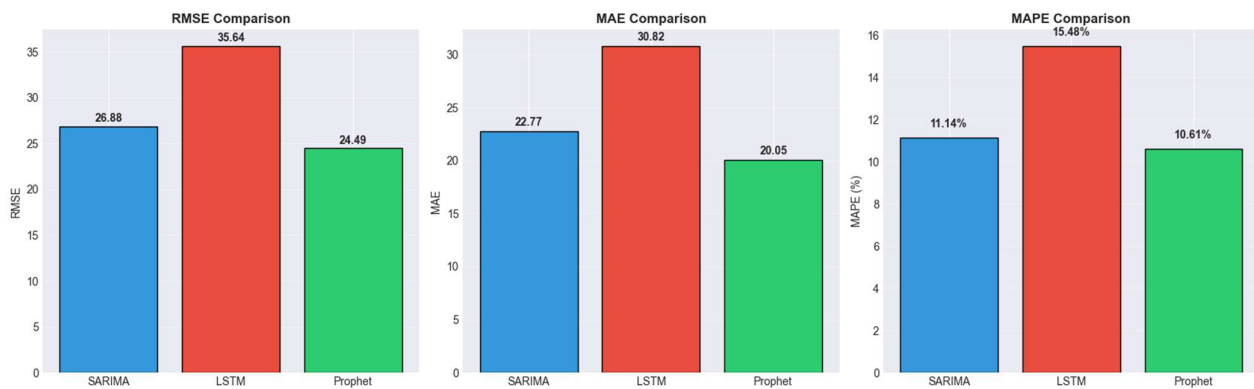
=== Model Comparison (Test Set) ===

Model	RMSE	MAE	MAPE (%)
SARIMA	26.876746	22.773062	11.137779
LSTM	35.641662	30.816630	15.482842
Prophet	24.486550	20.053772	10.608848

✓ Best RMSE: Prophet

✓ Best MAE: Prophet

✓ Best MAPE: Prophet



=== Normality Tests (Shapiro-Wilk) ===

H0: Errors are normally distributed

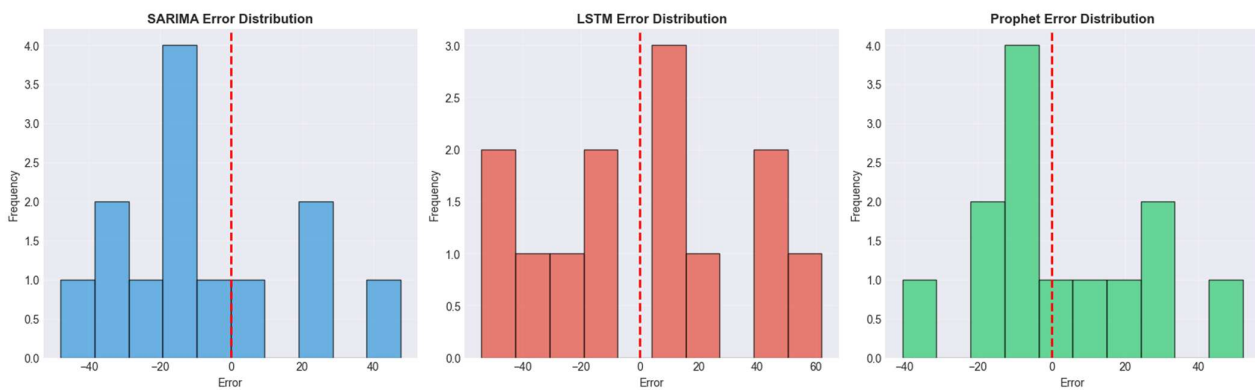
SARIMA: statistic=0.9613, p-value=0.7738

LSTM: statistic=0.9607, p-value=0.7643

Prophet: statistic=0.9692, p-value=0.8848

Interpretation: p-value > 0.05 suggests errors are normally distributed (good)

=== Error distributions comparison ===



1. DATA SUMMARY

City: Delhi

Total months: 67

Training period: 2015-01 to 2018-12 (48 months)

Test period: 2019-01 to 2020-01 (13 months)

AQI range: 67.2 - 474.8

2. MODEL PERFORMANCE (TEST SET)

Model	RMSE	MAE	MAPE (%)
SARIMA	26.876746	22.773062	11.137779
LSTM	35.641662	30.816630	15.482842
Prophet	24.486550	20.053772	10.608848

3. BEST MODELS BY METRIC

Lowest RMSE: Prophet (24.49)
Lowest MAE: Prophet (20.05)
Lowest MAPE: Prophet (10.61%)
...

=====

END OF REPORT

=====